Linear Algebra (MT-1004)

Homework#18

Sections: B,C & D

Instructor: Muhammad Ali

Question 1

Given two vectors

$$\mathbf{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \end{bmatrix},$$

find the area of the parallelogram formed by these vectors using the determinant.

Question 2

By what factor does the following transformation change the size of the box?

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - y \\ 2x + y \end{bmatrix}$$

Question 3

Find the determinant of the following matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 4 \\ 8 & 7 & 10 \end{bmatrix}$$

By using:

1. Cofactor Expansion method
2. Row operation

Question 4

Find the inverse of the given matrix using the determinant.

$$\begin{bmatrix} 5 & 6 & 0 & 2 \\ 2 & 1 & 3 & 9 \\ 7 & 0 & 5 & 8 \\ 4 & 8 & 1 & 4 \end{bmatrix}$$